

Modulation

Amplitude Modulation

Frequency modulation

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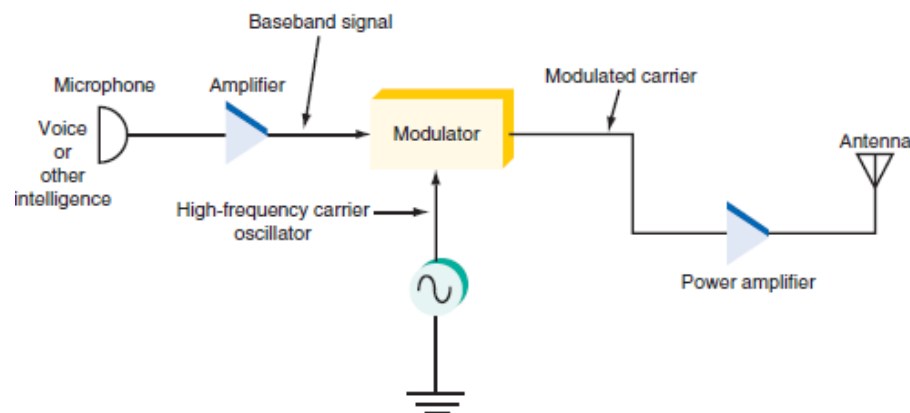
Phase Modulation

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Modulation

In the modulation process, the baseband voice, video, or digital signal modifies another, higher-frequency signal called the carrier, which is usually a sine wave. A sine wave carrier can be modified by the intelligence signal through amplitude modulation, frequency modulation, or phase modulation.

Modulation is a technique that changes the characteristics of the carrier frequency in accordance with the input signal.



where the modulation is performed at the transmit side and demodulation is performed at the receive side.

Need for modulation

- The baseband signals are incompatible for direct transmission. For such a signal, to travel longer distances, its strength has to be increased by modulating with a high frequency carrier wave, which doesn't affect the parameters of the modulating signal.
- The antenna used for transmission, had to be very large, if modulation was not introduced. The range of communication gets limited as the wave cannot travel to a distance without getting distorted.

Why the modulation is required in the communication system?

1- To avoid the interference

All information signals approximately in the same frequency leading to interference.

2- To get desired antenna height

The low frequency signal leads to impractical antenna height(large).

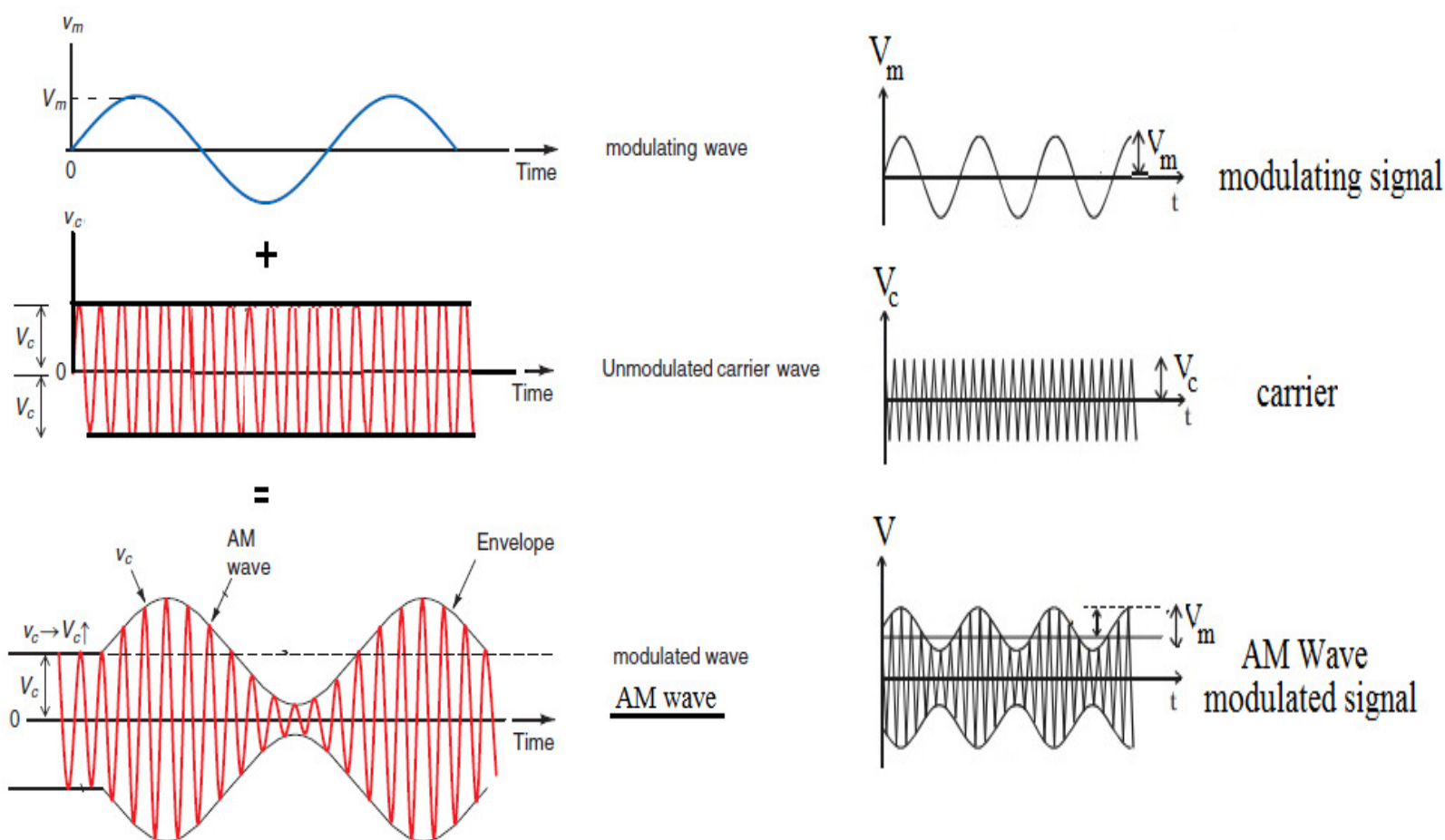
3- The ability of multiplexing

Left as assignment

AM modulation

In AM modulation the **amplitude** of the carrier is changed in accordance with modulating signal.

- ✓ Carrier: unmodulated signal
- ✓ Information signal: modulating signal
- ✓ Result of modulation: modulated signal



- The envelope of the modulating signal varies above and below the peak carrier amplitude.
- The amplitude of the modulating signal should be **less than or equal** the amplitude of the carrier.
- When the amplitude of the modulating signal is **greater than** the amplitude of the carrier, **distortion will occur**.

Modulating signal expression: $v_m = V_m \sin \omega_m t$ and **Carrier** expression: $v_c = V_c \sin \omega_c t$

Where

v_m is the **instantaneous value** of the **information signal**

V_m is the **peak amplitude** of the **information signal**

f_m is the **frequency** of the **information signal**

and

v_c is the **instantaneous value** of the **carrier signal**

V_c is the **peak amplitude** of the **carrier signal**

f_c is the **frequency** of the **carrier signal**

General AM expression

$$V_c = V_c \sin \omega_c t$$

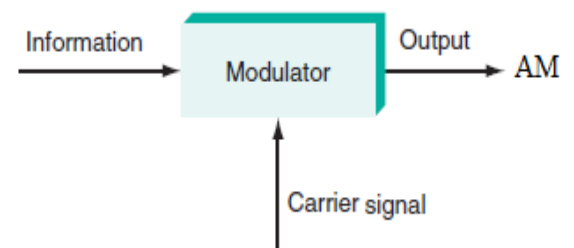
Carrier amplitude after modulation $\rightarrow V_{c,AM} = V_c + V_m = V_c + V_m \sin \omega_m t$

$$V_{Am} = V_{c,AM} \sin \omega_c t = [V_c + V_m \sin \omega_m t] \sin \omega_c t$$

$$V_{Am} = V_c \sin \omega_c t + V_m \sin \omega_c t \sin \omega_m t$$

Modulator

- A circuit that changes a lower frequency signal to a higher frequency signal is usually called a modulator. (The circuit used for producing AM)
- A circuit used to recover the original information signal from an AM wave is known as a detector or demodulator
- The modulator circuit has two inputs, one for the carrier and the second for the information one output for the AM wave
- Amplitude modulators compute the product of the carrier and modulating signals



Modulation index

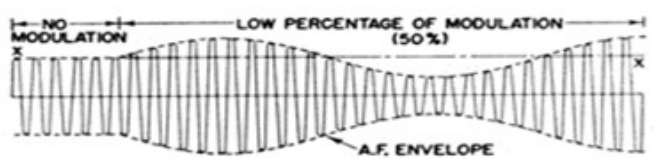
It is the ratio of the modulating signal amplitude to the carrier amplitude

$$m = \frac{V_m}{V_c}$$

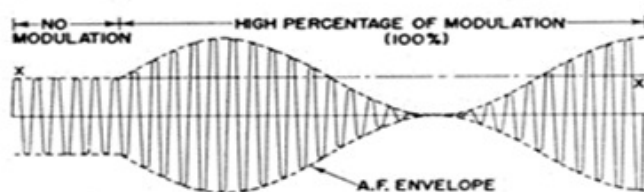
Over modulation

It is the situation at which modulation index is greater than 1 when $V_m > V_c$

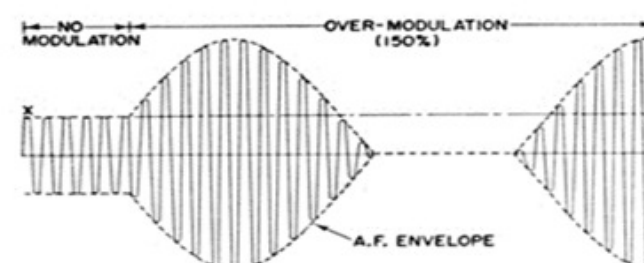
AM modulation degree depth



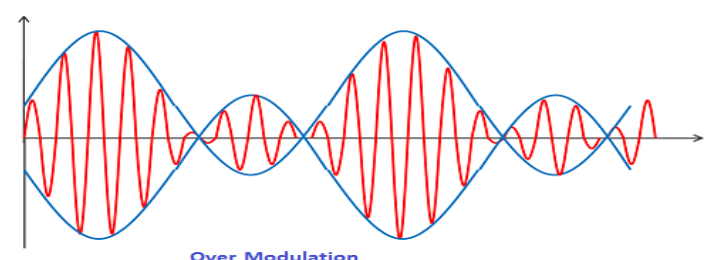
Ⓐ under modulation $m < 1$



Ⓑ maximum modulation $m = 1$



Ⓒ over modulation $m > 1$



Why the over modulation must be prevented in AM modulation?

- It causes **distortion** of the modulated AM signal.
- The original information signal becomes **unintelligible**.

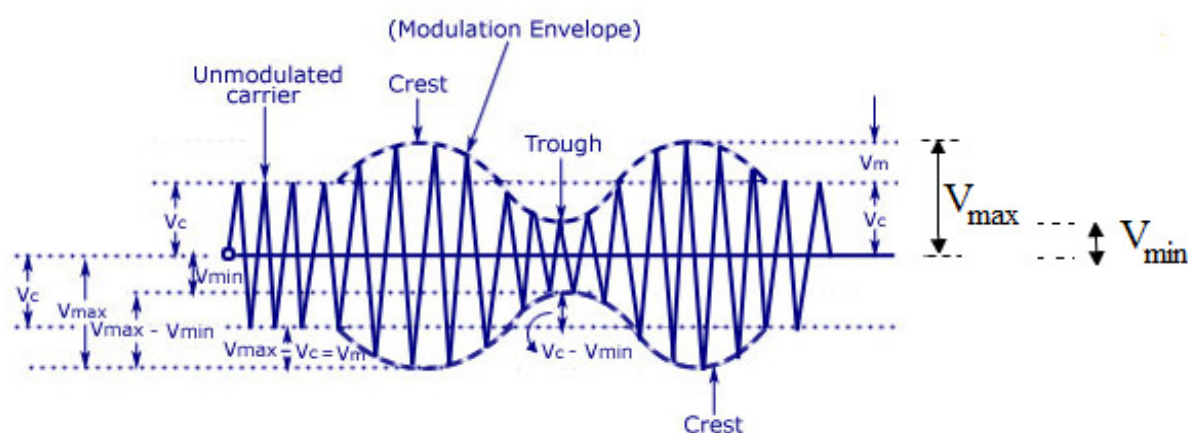
Notes

- Distortion of **voice transmissions** produces garbled, harsh, or unnatural sounds in the speaker.
- Distortion of **video transmissions** produces a scrambled and inaccurate picture on a TV screen
- The **ideal condition** for AM is when $V_m = V_c$, or $m = 1$, which gives **100 percent modulation**. This results in the **greatest output power** at the transmitter and the **greatest output voltage** at the receiver, with **no distortion**.

Automatic circuits (compression circuits)

A circuit that **prevents** over modulation and distortion.

- It **amplifies the lower level signals** and **compressing the higher level signals**. The result is a higher average power output level without over modulation.



AM wave

$$V_{\max} = V_c + V_m \quad \text{_____} \quad 1$$

$$V_{\min} = V_c - V_m \quad \text{_____} \quad 2$$

$$V_c = \frac{V_{\max} + V_{\min}}{2} \quad \text{and} \quad V_m = \frac{V_{\max} - V_{\min}}{2}$$

$$m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \quad \text{and} \quad m = \frac{V_m}{V_c}$$

Sidebands and the Frequency Domain

Sidebands expression

$$\therefore V_{Am} = V_c (\sin 2\pi f_c t) + (V_m \sin 2\pi f_c t) (\sin 2\pi f_m t)$$

$$V_{Am} = V_c (\sin 2\pi f_c t) + V_m (\sin 2\pi f_c t \times \sin 2\pi f_m t)$$

$$\therefore \sin \alpha \sin \beta = \frac{1}{2} [\cos (\alpha - \beta) - \cos (\alpha + \beta)]$$

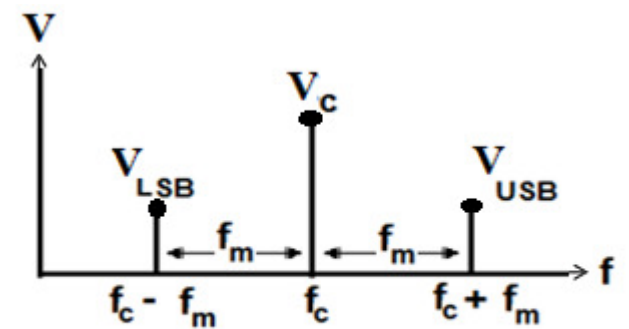
$$\therefore V_{Am} = V_c (\sin 2\pi f_c t) + V_m \times \frac{1}{2} [\cos 2\pi (f_c - f_m)t - \cos 2\pi (f_c + f_m)t]$$

$$V_{Am} = V_c (\sin 2\pi f_c t) + \frac{V_m}{2} \cos 2\pi (f_c - f_m)t - \frac{V_m}{2} \cos 2\pi (f_c + f_m)t$$

where the first term is the carrier, the second term, containing the difference $f_c - f_m$, is the lower sideband, and the third term, containing the sum $f_c + f_m$, is the upper sideband.

- **Upper** sideband frequency: $f_{USB} = f_c + f_m$
- **Lower** sideband frequency: $f_{LSB} = f_c - f_m$
- **BW = 2 f_m**

Where f_m is the **highest** modulating wave frequency



Frequency spectrum for AM wave

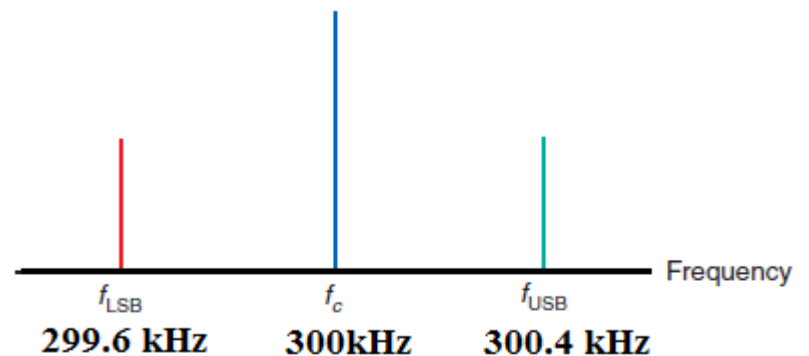
Example 1

Assume that a 400 Hz tone modulates a 300 kHz carrier. Sketch the AM frequency spectrum

- $f_m = 400 \text{ Hz} = 0.4 \text{ kHz}$, $f_c = 300 \text{ kHz}$

$$f_{USB} = f_c + f_m = 300 + 0.4 = 300.4 \text{ kHz}$$

$$f_{LSB} = f_c - f_m = 300 - 0.4 = 299.6 \text{ kHz}$$



Example 2

A standard AM broadcast station is allowed to transmit modulating frequencies up to 5 kHz. If the AM station is transmitting on a frequency of 980 kHz, compute the **maximum** and **minimum** upper and lower sidebands and the total **bandwidth** occupied by the AM station.

Solution

- $f_m = 5 \text{ kHz}$, $f_c = 980 \text{ kHz}$

$$f_{USB} = f_c + f_m = 980 + 5 = 985 \text{ kHz}$$

$$f_{LSB} = f_c - f_m = 980 - 5 = 975 \text{ kHz}$$

$$BW = 2 f_m = 2 \times 5 = 10 \text{ kHz}$$

Example 3

In AM modulation a carrier of 1MHz frequency and 10 V peak voltage is modulated by 10 kHz and 5 V modulating wave.

- Write an expression for modulated AM wave.
- Draw the frequency spectrum of AM wave.

Solution

- $V_c = 10 \text{ V}$, $f_c = 1 \text{ MHz} = 10^6 \text{ Hz}$
- $V_m = 5 \text{ V}$, $f_m = 10 \text{ kHz} = 10 \times 10^3 \text{ Hz} = 0.01 \times 10^6 \text{ Hz}$

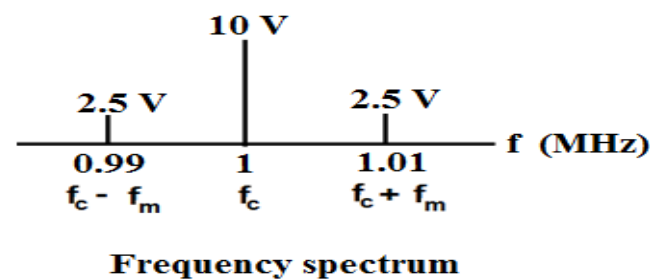
a)

- $f_c + f_m = 10^6 + 0.01 \times 10^6 = 1.01 \times 10^6 \text{ Hz} = 1.01 \text{ MHz}$
- $f_c - f_m = 10^6 - 0.01 \times 10^6 = 0.99 \times 10^6 \text{ Hz} = 0.99 \text{ MHz}$

$$V_{Am} = V_c \cos 2\pi f_c t + \frac{V_m}{2} \cos 2\pi(f_c - f_m)t - \frac{V_m}{2} \cos 2\pi(f_c + f_m)t$$

$$V_{Am} = 10 \cos 2\pi(10^6)t + 2.5 \cos 2\pi(0.99 \times 10^6)t - 2.5 \cos 2\pi(1.01 \times 10^6)t$$

b)

**AM Power**

In radio transmission, the AM signal is amplified by a power amplifier and fed to the antenna with characteristic **impedance** that is ideally, but not necessarily, almost **pure resistance**.

Power in AM wave

$$\text{➤ } P = \frac{V_{rms}^2}{R} \text{ and } V_{rms} = \frac{V_c}{\sqrt{2}}$$

$$\text{➤ } \textbf{Carrier Power} : P_c = \frac{\left(\frac{V_c}{\sqrt{2}}\right)^2}{R} = \frac{V_c^2}{2R}$$

$$\text{➤ } \textbf{For only one side band} : P_{USB} = P_{LSB} = \frac{\left(\frac{V_m}{2\sqrt{2}}\right)^2}{R} = \frac{V_m^2}{8R}$$

$$\text{➤ } \textbf{Side band power in terms of carrier power} : P_{USB} = P_{LSB} = \frac{m^2}{4} P_c$$

$$\text{➤ } \textbf{Double or both side bands} : P_{SB} = \frac{m^2}{2} P_c$$

Total AM power

$$P_t = P_c + P_{USB} + P_{LSB} = P_c + \frac{m^2}{4} P_c + \frac{m^2}{4} P_c$$

$$\text{➤ } P_t = P_c \left(1 + \frac{m^2}{2}\right)$$

$$\text{➤ } \text{Maximum power of AM occurs when } m = 1: P_t = P_c \left(1 + \frac{1^2}{2}\right) = 1.5 P_c$$

Simultaneous modulation (more than one sine wave or modulating signal):

$$\triangleright m_t = \sqrt{m_1^2 + m_2^2 + m_3^2 + \dots}$$

Example 4

An AM transmitter has a carrier power of 30 W. The percentage of modulation is 85 percent. Calculate

- the total power
- the power in one sideband.

Solution

$$\blacksquare P_c = 30 \text{ W}, m = \frac{85}{100} = 0.85$$

$$\text{a) } P_t = P_c \left(1 + \frac{m^2}{2}\right) = 30 \left(1 + \frac{0.85^2}{2}\right) = 40.8 \text{ W}$$

$$\text{b) } P_{\text{USB}} = P_{\text{LSB}} = \frac{m^2}{4} P_c = \frac{0.85^2}{4} \times 30 = 5.4 \text{ W}$$

Example 5

for a standard AM transmission $v_c = 10 \cos(8\pi \times 10^5 t)$ and $v_m = 6 \cos(\pi \times 10^5 t)$.

The transmitter feeds a resistive load of 10 k Ω . Determine

- The transmitted total power.
- The AM modulated waveform bandwidth.
- If the total power was reduced by 10% determine the value of the modulation index.

Solution

$$\blacksquare V_c = 10 \text{ V}, V_m = 6 \text{ V}, R = 10 \text{ k}\Omega = 10000 \Omega.$$

a)

$$P_c = \frac{V_c^2}{2R} = \frac{10^2}{2 \times 10000} = 0.005 \text{ W}$$

$$m = \frac{V_m}{V_c} = \frac{6}{10} = 0.6 \quad \text{and} \quad P_t = P_c \left(1 + \frac{m^2}{2}\right) = 0.005 \left(1 + \frac{0.6^2}{2}\right) = 0.0059 \text{ W}.$$

b)

$$\omega_m = 2\pi f \quad \longrightarrow \quad f = \frac{\omega_m}{2\pi} = \frac{\pi \times 10^5}{2\pi} = 50 \times 10^3 \text{ Hz} = 50 \text{ kHz}$$

$$\text{Bandwidth of AM} = 2 f_m = 2 \times 50 \text{ kHz} = 100 \text{ kHz}$$

c)

$$P_t \text{ reduced by } 10\% = 0.1: P_t = (1 - 0.1) \times 0.0059 = 0.00531 \text{ W}$$

$$\frac{P_t}{P_c} = \left(1 + \frac{m^2}{2}\right) \quad \longrightarrow \quad m^2 = 2 \left(\frac{P_t}{P_c} - 1\right) = 2 \left(\frac{0.00531}{0.005} - 1\right) = 0.124$$

$$m = \sqrt{0.124} = 0.352 = 35.2 \%$$

Example 6

For a standard AM transmission, the maximum peak – to – peak voltage is 150 V and the minimum peak – to – peak voltage is 50 V. What is the percentage of the total power to the unmodulated carrier power?

Solution

For peak to peak: $V_{\max (P-P)} = 150 \text{ V}$, $V_{\min (P-P)} = 50 \text{ V}$

$$\text{➤ For peak : } V_{\max} = \frac{150}{2} = 75 \text{ V} , V_{\min} = \frac{50}{2} = 25 \text{ V}$$

$$V_c = \frac{V_{\max} + V_{\min}}{2} = \frac{75 + 25}{2} = 50 \text{ V}.$$

$$V_m = \frac{V_{\max} - V_{\min}}{2} = \frac{75 - 25}{2} = 25 \text{ V}.$$

$$V_{\text{USB}} = V_{\text{LSB}} = \frac{V_m}{2} = \frac{25}{2} = 12.5 \text{ V} .$$

$$m_a = \frac{V_m}{V_c} = \frac{25}{50} = 0.5 = 50\%$$

$$\text{Total power to the unmodulated carrier power : } P_t = P_c \left(1 + \frac{m^2}{2}\right) \quad \longrightarrow \quad \frac{P_t}{P_c} = \left(1 + \frac{m^2}{2}\right)$$

$$\frac{P_t}{P_c} = \left(1 + \frac{0.5^2}{2}\right) = 1.125 = 112.5\%$$

Example 7

It is given that 10 V carrier is modulated to depth 50 percent and 50 Ω antenna resistance. Find the total power and each side band power.

Solution

$$\blacksquare V_c = 10 \text{ V} , R = 50 \Omega , m = 50 \% = 0.5$$

$$P_c = \frac{V_c^2}{2R} = \frac{10^2}{2 \times 50} = 1 \text{ W}$$

$$\text{➤ } P_t = P_c \left(1 + \frac{m^2}{2}\right) = 1 \left(1 + \frac{0.5^2}{2}\right) = 1.125 \text{ W}$$

$$\text{➤ } P_{\text{USB}} = P_{\text{LSB}} = \frac{m^2}{4} P_c = \frac{0.5^2}{4} \times 1 = 0.0625 \text{ W}$$

Example 8

The antenna current of an AM broadcast transmitter, modulate to a depth of 40 percent by an audio sine wave is 11 A .it increased to 12A as a result of simultaneous modulation by another audio sine wave. What is the modulation index due to this second wave?

Solution

$$\blacksquare m_1 = 40\% = 0.4 , I_{t1} = 11 \text{ A (first modulation)} , I_t = 12 \text{ A}$$

$$\text{➤ } I_{t1} = I_c \sqrt{1 + \frac{m_1^2}{2}} \quad \longrightarrow \quad I_c = \frac{I_{t1}}{\sqrt{1 + \frac{m_1^2}{2}}} = \frac{11}{\sqrt{1 + \frac{0.4^2}{2}}} = 10.58 \text{ A}$$

$$I_t = I_c \sqrt{1 + \frac{m_t^2}{2}} \quad \longrightarrow \quad \left(\frac{I_t}{I_c}\right)^2 = \left(1 + \frac{m_t^2}{2}\right) \quad \longrightarrow \quad \left(\frac{12}{10.58}\right)^2 = \left(1 + \frac{m_t^2}{2}\right)$$

$$\triangleright m_t = 0.757$$

$$m_t = \sqrt{m_1^2 + m_2^2} \longrightarrow m_2 = \sqrt{m_t^2 - m_1^2} = \sqrt{0.757^2 - 0.4^2} = 0.643$$

AM transmission

To reconstruct the information signal at the receiver side by the demodulation process, it is not necessary to transmit all of the USB, LSB and the carrier at the transmitted side, therefore to save the power at the transmitter, we have more than one techniques are used.

In amplitude modulation

- Two-thirds of the transmitted power is in the carrier, which has no information.
- The information is only within the sidebands.

1) Double Side Bands Full Carriers (DSBFC):

- Both of side bands (USB + LSB) and carrier are transmitted.
- simple detected circuit is needed at the receiver.
- waste the bandwidth.
- large power used.

2) Double Side Bands Suppressed Carriers (DSBSC):

- Both of side bands USB + LSB are transmitted.
- Complex detected circuit is needed at the receiver.
- Less power is used than DSBFC.
- Waste the bandwidth.

One way to improve the efficiency of amplitude modulation

- Suppress the carrier and eliminate one sideband.

3) Single Side Band (SSB):

- Only one of the side Band USB or LSB is transmitted.
- More complex detected circuit is needed at the receiver.
- Low power is needed and much power is saving (because only one side band is transmitted).

Saving power in AM

$$P_t = P_c \left(1 + \frac{m^2}{2}\right)$$

For $m = 1$:

$$\triangleright \text{Total power: } P_t = P_c \left(1 + \frac{m_a^2}{2}\right) = P_c \left(1 + \frac{1^2}{2}\right) = 1.5 P_c$$

$$\triangleright \text{Both side bands power (DSB): } P_{DSB} = \frac{m_a^2}{2} P_c = \frac{1^2}{2} P_c = 0.5 P_c$$

$$\triangleright \text{one side band power: } P_{USB} = P_{LSB} = \frac{m_a^2}{4} P_c = \frac{1^2}{4} P_c = 0.25 P_c$$

In DSBSC two third a power is saved but in SSB more saving power is increased.

Example 9

Calculate the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of (a) 100 percent and (b) 50 percent.

Solution

c) $m_a = 100\% = 1$

➤ Total power: $P_t = 1.5 P_c$

➤ one side band power: $P_{USB} = P_{LSB} = \frac{m_a^2}{4} P_c = \frac{1^2}{4} P_c = 0.25 P_c$

$$\text{Saving power} = \frac{1.5 - 0.25}{1.5} = 0.833 = 83.3\% \text{ of the total power is saved}$$

d) $m_a = 50\% = 0.5$

➤ Total power: $P_t = P_c \left(1 + \frac{m_a^2}{2}\right) = P_c \left(1 + \frac{0.5^2}{2}\right) = 1.125 P_c$

➤ one side band power: $P_{USB} = P_{LSB} = \frac{m_a^2}{4} P_c = \frac{0.5^2}{4} P_c = 0.0625 P_c$

$$\text{Saving power} = \frac{1.125 - 0.0625}{1.125} = 0.944 = 94.4\% \text{ of the total power is saved}$$

Advantages of AM modulation

- It is simple to implement
- It can be demodulated using a circuit (detector) of very few components.
- AM receivers are very cheap as no specialized components are needed.

Disadvantages of AM modulation:

- An amplitude modulation signal is not efficient in terms of its power usage.
- It is not efficient in terms of its use of bandwidth, requiring a bandwidth equal to twice that of the highest modulating signal frequency
- An amplitude modulation signal is prone to high levels of noise because most noise is amplitude based and obviously AM detectors are sensitive to it.

Despite its inefficiency, AM is **still widely used** because it is **simple** and **effective**. It is used in

- 1- AM radio broadcasting.
- 2- TV broadcasting.
- 4- Aircraft tower communication.

Conceptual questions

- 1- Compare between classical AM modulation and single sideband modulation.
- 2- Mention two disadvantages sideband modulations.
- 3- List down five applications involved AM modulation technique.

Problems**Q₁**

An antenna has an impedance of 40 Ω . An unmodulated AM signal produces a current of 4.8 A. The modulation is 90 percent. Calculate

- (a) the carrier power
- (b) the sidebands power and total power
- (c) If the antenna current change from 4.8A unmodulated to 5.1 A. What is the percentage of modulation?

Ans. a) 921.6 W, b) 373.3 W, 1295 W and c) 51 %

Q₂

A 40 V peak to peak 500 kHz carrier is modulated by a 30 V peak to peak sine wave. Determine

- a) The maximum and the minimum voltages of the modulated waveform.
- b) The amplitude of the side frequencies.
- c) The modulation index.
- d) The total power to the unmodulated carrier power.
- e) The modulation index if the carrier frequency increased to 600 kHz.

Ans. a) 35 V, 5 V, b) 7.5 V , c) 75 % , d) 128.125 % , and e) constant

Q₃

The carrier for a standard AM transmission is given by $v_c = 7 \cos(2\pi \times 10^5 t)$ and modulating wave is given by $v_m = 4 \cos(2\pi \times 500 t)$.

- a) Draw the frequency spectrum.
- b) Determine the bandwidth of AM signal.

Ans. a) 100.5 kHz , 99.5 kHz and b) 1 kHz

Q₄

It is given that 10 V rms value of the carrier is modulated to depth 50 percent and 50 Ω antenna resistance. Find the total power and each side band power.

Ans. 2.25 W and 0.125 W

Q₅

The carrier for a standard AM transmission is given by

$v_c = 10 \cos(8\pi \times 10^5 t)$ and one side frequency (Lower side band) for sinusoidal modulation is given by $v_{LSB} = 3 \cos(\pi \times 10^5 t)$.

The transmitter feeds a resistive load of 10 k Ω . Determine

- a) The transmitted total power.
- b) The modulated waveform bandwidth.
- c) If the total power was reduced by 10% determine the value of the modulation index.

Ans. a) 5.9 mW , b) 700 kHz , 35.2 %

Frequency modulation

In FM the frequency of the carrier is changed in accordance with modulating signal while its amplitude is kept constant.

FM frequency representation

- Let the modulating signal voltage represented by $v_m = V_m \cos \omega_m t = V_m \cos 2\pi f_m t$
- Let the carrier voltage represented by $v_c = V_c \cos \omega_c t = V_c \cos 2\pi f_c t$

The carrier frequency will change by means of the modulating signal, therefore

$$v_c = V_c \cos 2\pi [f_c + k_f V_m \cos \omega_m t] t$$

Let the FM frequency is $f = f_c + k_f V_m \cos \omega_m t$

The maximum frequency deviation occurred when $\cos \omega_m = \pm 1$

$$f = f_c \pm k_f V_m \longrightarrow f = f_c \pm \delta$$

- Where δ is the frequency deviation $\delta = k_f V_m$, and k_f is proportionality constant in Hz /Volt.

FM signal representation

Let $\Phi = \omega t$

$$d\Phi = \omega dt \longrightarrow \Phi = \int \omega dt, \quad \text{note that } \omega = 2\pi [f_c + \delta \cos 2\pi f_m t]$$

$$\Phi = \int \omega dt = \int 2\pi [f_c + \delta \cos 2\pi f_m t] dt = \int 2\pi f_c dt + \int 2\pi \delta \cos 2\pi f_m t dt$$

$$\Phi = 2\pi f_c t + \frac{2\pi \delta}{2\pi f_m} \sin 2\pi f_m t \longrightarrow \Phi = [\omega_c t + \frac{\delta}{f_m} \sin \omega_m t]$$

$$v_{FM} = V_c \cos \Phi \longrightarrow v_{FM} = V_c \cos [\omega_c t + m_f \sin \omega_m t]$$

Where m_f is the modulation index for FM : $m_f = \frac{\delta}{f_m}$

- Frequency deviation** is the maximum deviation of the instantaneous FM frequency from the carrier frequency.
- In FM the **modulation index** is defined as the ratio of the maximum frequency deviation to the modulating frequency.
- Maximum frequency deviation depends on the amplitude of the modulating signal.*
- FM modulation index is proportional to the modulating signal amplitude and inversely proportional to the modulating signal frequency.*

Example 10

In FM transmission it is given that $f_m = 500 \text{ Hz}$, $V_m = 2.4 \text{ V}$ and the deviation is 4.8 kHz , if the modulating signal is 7.2 V . What is the new deviation, also find the modulation index in each case?

Solution

- $f_m = 500 \text{ Hz} = 0.5 \text{ kHz}$, $V_m = 2.4 \text{ V}$, $\delta = 4.8 \text{ kHz}$

$$k_f = \frac{\delta}{V_m} = \frac{4.8 \text{ k}}{2.4} = 2 \text{ kHz/V}$$

$$m_{f1} = \frac{\delta_1}{f_m} = \frac{4.8 \text{ k}}{0.5 \text{ k}} = 9.6$$

- for $V_m = 7.2 \text{ V}$

$$\delta = k_f V_m = 2 \text{ k} \times 7.2 = 14.4 \text{ kHz}$$

$$m_{f2} = \frac{\delta_2}{f_m} = \frac{14.4 \text{ k}}{0.5 \text{ k}} = 28.8$$

Example 11

Find the carrier and the modulating frequencies, the modulation index and the maximum deviation of the FM wave represented by $v_{FM} = 12 \cos [6 \times 10^8 t + 5 \sin 1250 t]$. What power will this FM wave dissipate in a 10Ω resistor?

Solution

- $V_c = 12 \text{ V}$, $\omega_c = 6 \times 10^8 \text{ rad/s}$, $\omega_m = 1250 \text{ rad/s}$, $m_f = 5$, $R = 10 \Omega$

- $\omega_c = 2\pi f \quad \longrightarrow \quad f_c = \frac{\omega_c}{2\pi} = \frac{6 \times 10^8}{2\pi} = 95.5 \times 10^6 \text{ Hz} = 95.5 \text{ MHz}$

- $\omega_m = 2\pi f \quad \longrightarrow \quad f_m = \frac{\omega_m}{2\pi} = \frac{1250}{2\pi} = 198.9 = 199 \text{ Hz}$

- $\delta = m_f f_m = 5 \times 199 = 995 \text{ Hz}$

- $P_c = \frac{V_c^2}{2R} = \frac{12^2}{2 \times 10} = 7.2 \text{ W}$

PM modulation

In PM the phase of the carrier is changed in accordance with modulating signal.

PM signal representation

$$V_{PM} = V_c \cos [\omega_c t + m_p \sin \omega_m t]$$

- Where m_p is the modulation index for PM : $m_p = k_p V_m$, and k_p is proportionality constant in radian/Volt.
 - *PM modulation index is proportional to the modulating signal amplitude and it is independent of the modulating signal frequency.*

$$\phi = k_p V_m \quad \longrightarrow \quad k_p = \frac{\phi}{V_m} \quad \text{in rad/volt}$$

Example 12

A 25-MHz carrier is modulated by a 400-Hz audio sine wave. If the carrier voltage is 4 V and the maximum deviation is 10 kHz, write the equations of this modulated wave for

(a) FM and (b) PM

If the modulating frequency is now changed to 2 kHz, all else remaining constant write new equations for

(b) FM and (d) PM

Solution

$$V_c = 4 \text{ V} , f_c = 25 \text{ MHz} = 25 \times 10^6 \text{ Hz} , f_m = 400 \text{ Hz} = 0.4 \text{ kHz} , m_f = 5, R = 10 \Omega$$

- $\omega_c = 2\pi \times f_c = 2\pi \times 25 \times 10^6 = 50\pi \times 10^6 \text{ rad/s}$
- $\omega_m = 2\pi \times f_m = 2\pi \times 400 = 800\pi \text{ rad/s}$

$$\text{a) } m_f = \frac{\delta}{f_m} = \frac{10 \text{ k}}{0.4 \text{ k}} = 25 \quad \text{and} \quad m_p = 25$$

$$\text{➤ FM: } V(t) = V_c \cos [\omega_c t + m_f \sin \omega_m t] = 4 \cos [50\pi \times 10^6 t + 25 \sin 800\pi t]$$

$$\text{➤ PM: } V(t) = V_c \cos [\omega_c t + m_p \sin \omega_m t] = 4 \cos [50\pi \times 10^6 t + 25 \sin 800\pi t]$$

$$\text{b) } f_m = 2 \text{ kHz} = 2 \times 10^3 \text{ Hz}$$

$$\bullet \omega_m = 2\pi \times f_m = 2\pi \times 2 \times 10^3 = 4\pi \times 10^3 \text{ rad/s}$$

$$m_f = \frac{\delta}{f_m} = \frac{10 \text{ k}}{2 \text{ k}} = 5 \quad \text{and} \quad m_p = 25 \quad (m_p \text{ is independent of the } f_m)$$

$$\text{➤ FM: } v_{FM} = V_c \cos [\omega_c t + m_f \sin \omega_m t] = 4 \cos [50\pi \times 10^6 t + 5 \sin 4\pi \times 10^3 t]$$

$$\text{➤ PM: } v_{PM} = V_c \cos [\omega_c t + m_p \sin \omega_m t] = 4 \cos [50\pi \times 10^6 t + 25 \sin 4\pi \times 10^3 t]$$

Example 13

Carrier signal voltage is 7 V and the frequency is 5 MHz

Modulating signal voltage is 2 V and the frequency is 5 kHz

The frequency deviation is 4 kHz and the antenna resistance is 10 Ω

- Write an expression for FM and PM voltages.
 - If the modulating voltage increases to 4 V. Write new expression for FM and PM.
 - If the modulating frequency increases to 10 kHz. Write new expression for FM and PM.
- In each case determine the modulation indices and the transmitted power.

Solution

$V_c = 7 \text{ V}$, $f_c = 5 \text{ MHz} = 5 \times 10^6 \text{ Hz}$, $f_m = 5 \text{ kHz} = 5 \times 10^3 \text{ Hz}$, $\delta = 4 \text{ kHz}$, $R = 10 \Omega$

- $\omega_c = 2\pi \times f_c = 2\pi \times 5 \times 10^6 = \pi \times 10^7 \text{ rad/s}$
- $\omega_m = 2\pi \times f_m = 2\pi \times 5 \times 10^3 = \pi \times 10^4 \text{ rad/s}$

$$\triangleright k_f = \frac{\delta}{V_m} = \frac{4 \text{ k}}{2} = 2 \text{ k Hz/V}$$

$$m_f = \frac{\delta}{f_m} = \frac{4 \text{ k}}{5 \text{ k}} = 0.8 \quad \text{and} \quad m_p = 0.8, \quad \phi = 0.8 \text{ rad}, \quad k_p = \frac{\phi}{V_m} = \frac{0.8}{2} = 0.4 \text{ rad/V}$$

a)

$$\triangleright \text{FM: } v_{\text{FM}} = V_c \cos [\omega_c t + m_f \sin \omega_m t] = 7 \cos [\pi \times 10^7 t + 0.8 \sin \pi \times 10^4 t]$$

$$\triangleright \text{PM: } v_{\text{PM}} = V_c \cos [\omega_c t + m_p \sin \omega_m t] = 7 \cos [\pi \times 10^7 t + 0.8 \sin \pi \times 10^4 t]$$

$$\triangleright P_c = \frac{V_c^2}{2R} = \frac{7^2}{2 \times 10} = 2.45 \text{ W}$$

b) For $V_m = 4 \text{ V}$

- $\delta = k_f V_m = 2 \text{ k} \times 4 = 8 \text{ kHz}$

- $m_f = \frac{\delta}{f_m} = \frac{8 \text{ k}}{5 \text{ k}} = 1.6$

- $m_p = k_p V_m = 0.4 \times 4 = 1.6$

$$\triangleright \text{FM: } v_{\text{FM}} = V_c \cos [\omega_c t + m_f \sin \omega_m t] = 7 \cos [\pi \times 10^7 t + 1.6 \sin \pi \times 10^4 t]$$

$$\triangleright \text{PM: } v_{\text{PM}} = V_c \cos [\omega_c t + m_p \sin \omega_m t] = 7 \cos [\pi \times 10^7 t + 1.6 \sin \pi \times 10^4 t]$$

$$\triangleright P_c = \frac{V_c^2}{2R} = \frac{7^2}{2 \times 10} = 2.45 \text{ W}$$

c) For $f_m = 10 \text{ kHz} = 10 \times 10^3 \text{ Hz}$

$$\omega_m = 2\pi \times f_m = 2\pi \times 10 \times 10^3 = 2\pi \times 10^4 \text{ rad/s}$$

- $m_f = \frac{\delta}{f_m} = \frac{4 \text{ k}}{10 \text{ k}} = 0.4$

- $m_p = 0.8$ (m_p is independent of the f_m)

➤ FM: $v_{FM} = V_c \cos [\omega_c t + m_f \sin \omega_m t] = 7 \cos [\pi \times 10^7 t + 0.4 \sin 2\pi \times 10^4 t]$

➤ PM: $v_{PM} = V_c \cos [\omega_c t + m_p \sin \omega_m t] = 7 \cos [\pi \times 10^7 t + 0.8 \sin 2\pi \times 10^4 t]$

➤ $P_c = \frac{V_c^2}{2R} = \frac{7^2}{2 \times 10} = 2.45 \text{ W}$

Comparison between FM and AM

FM	AM
Carrier frequency is changed	Carrier amplitude is changed
infinite number of sidebands	Two sidebands
Useful using of the transmitted power , and there is no wastage of power	most of the transmitted power in the carrier which contains no information., wastage of power
Less effects by noise since the information is in carrier frequency	Higher effects by noise, since the information is in the carrier amplitude
Noise in FM can be reduced by amplitude limiters at the receiver to remove the variations caused by noise.	AM is more noisy since the AM receivers do not have amplitude limiters
Transmission and reception equipment is expensive as the circuitry is complex	Transmission and reception equipments are not that expensive since the circuitry is relatively simple
Broadcast operates in the upper VHF and UHF range, where noise effects are minimal	Broadcast operates in the medium frequency (MF) and high frequency (HF) where noise effects are higher

Disadvantages of FM

- A much wider channel is required by FM, up to 10 times as large as that needed by AM.
- FM transmitting and receiving equipment tends to be more complex, particularly for modulation and demodulation.
- The area of reception for FM is much smaller than for AM. It is a disadvantage for FM mobile communications over a wide area.

References:

1- **Principle of Electronic Communication Systems** by Louis E. Frenzel Jr ,4th edition.

2- **Electronics Communication Systems** by George Kennedy , 4th edition.